

Red Hat Lightspeed 1-latest

Deploying and managing RHEL systems in hybrid clouds

Deploying and managing your customized RHEL system images in hybrid clouds

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Abstract

Red Hat Lightspeed enables you to perform and manage multiple services as part of one larger, connected workflow. You can define your third-party content source, create customized images, use your preferred system by setting customizations in the image, and launch the image to the target public or private cloud environments, such as Amazon Web Services and Microsoft Azure. You can monitor the system infrastructure you have created, and create and apply patches when needed. You can also report and audit any issues you find in your system infrastructure to improve security and stability.

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CHAPTER 1. WORKFLOW FOR DEPLOYING AND MANAGING RHEL SYSTEMS IN OPEN HYBRID CLOUD ENVIRONMENTS

Use Red Hat Lightspeed to launch and manage your customized RHEL systems images in the open hybrid cloud environments.

You can use Red Hat Lightspeed to deploy and manage the RHEL systems in the following clouds:

- Amazon Web Services
- Microsoft Azure
- Google Cloud

You can access the Red Hat Lightspeed with your Red Hat account, a RHEL subscription, and an activation key. No additional SKUs are required. See Creating an activation key.

As part of the deployment, by using Red Hat Lightspeed, you can follow an end-to-end workflow to:

1. Define or use existing repositories.
2. Create a blueprint to build customized images .

In addition, you can manage your systems by granting users access, monitoring the inventory of your system infrastructure, and applying patches to improve security and stability.

CHAPTER 2. MANAGE REPOSITORIES TO BUILD YOUR CUSTOMIZED OPERATING SYSTEMS

You can define your customized repositories with third-party content without having to manage their lifecycle. You can use your third-party content to build an image, and when you launch that image to the public cloud environment, you can use those repositories with the **dnf** tool.

2.1. ADD A CUSTOM REPOSITORY

Define your repository to be able to add packages from this repository to your customized images.

Prerequisites

- You have a RHEL subscription.
- You have administrator access to the Red Hat Hybrid Cloud Console web user interface or **repository administrator** role.
- You have the URL link to your repository content.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Content → Repositories**. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console.
2. Click **Add repositories**. The **Add custom repositories** wizard opens.
3. In the **Name** field enter a name for your custom repository.
4. In the **Repository type**, select:

   Snapshotting

   Enables creating a daily snapshot of this repository if this repository has updates since the last snapshot. That enables you to create Image Blueprints with consistent repository content.

   Introspect only

   Disables snapshots for this repository.

   Upload

   Enables uploading packages to your custom repository. The file must have an **rpm** extension.
5. If you selected **Snapshotting** or **Introspect only**, in the **URL** field, enter the URL to your repository.
6. Optional: In the **Restrict architecture** drop-down menu, select an architecture. You can allow all the architectures or restrict it to your system architecture to prevent incorrect repositories availability.
7. Optional: In the **Restrict OS version** drop-down menu, select an operating system (OS). You can allow all the RHEL versions or select one for your system version to prevent incorrect repositories being available.

8. Optional: Disable **Modularity filtering** option. When the **Modularity filtering** option is disabled, you can update packages in this repository even if the packages are part of a module.

9. Optional: In the **GPG key** field, upload the **.txt** file with a GPG key or paste the URL or value of an existing GPG key. You can use the GPG key to verify the signed packages of a repository. If you do not provide the GPG key for a repository, your system cannot perform the verification.

10. If you selected **Snapshotting** or **Introspect only**, click **Save**. The Red Hat Hybrid Cloud Console validates the project status. If your repository is in the **Invalid** status, check the repository URL that you added. For details about the repository status, see the Repository status section.

11. If you selected **Upload**:

    a. Click **Save and upload content**. The **Upload content** window opens.

    b. Click **Upload**, select the **rpm** files you want to upload, and click **Open**.

    c. Click **Confirm changes** when your file uploading is complete.

Verification

- On the **Repositories** page, select the **Custom** tab, and verify that the repository you added is listed. Note, that the repositories from the **EPEL** and the **Red Hat** tabs are available to your custom images by default.

2.2. CONFIGURE A CUSTOM REPOSITORY

You can configure a custom repository when you need to update information for that repository.

Prerequisites

- You have a RHEL subscription.
- You have administrator access to the Red Hat Hybrid Cloud Console web user interface or **repository administrator** role.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Content → Repositories**. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console.

2. Find a repository you want to configure and click **Edit** in the **Options** menu.

3. In the **Edit custom repository** wizard, edit the information you need. Click **Save changes**.

2.3. REMOVE A CUSTOM REPOSITORY

When you no longer need a custom repository you can delete it.

Prerequisites

- You have a RHEL subscription.

- You have administrator access to the Red Hat Hybrid Cloud Console web user interface or **repository administrator** role.

Procedure

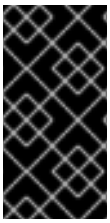
1. Access the Hybrid Cloud Console, and navigate to **Content → Repositories**. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console.
2. Find a repository to delete and click **Delete** in the **Options** menu.

Verification

- Open the list of custom repositories, and verify that the repository no longer exists.

2.4. REMOVE SNAPSHOTS FROM A REPOSITORY

You can delete snapshots from your custom repository to avoid broken functionality or security vulnerabilities that the old content might introduce.



IMPORTANT

Snapshots get removed automatically after 365 days unless there is no newer snapshot of this repository. If a repository has multiple snapshots and the snapshot for removal is used in a content template, this snapshot will be replaced with the newer snapshot in the content template.

Prerequisites

- You have a RHEL subscription.
- You have administrator access to the Red Hat Hybrid Cloud Console web user interface or **repository administrator** role.
- You have added a custom repository. See Adding a custom repository.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Content → Repositories**. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console.
2. In the **Your repositories** tab, find the repository containing the snapshot you want to remove, and click **View all snapshots** in the **Option** menu.
3. In the **Snapshot** window, select all snapshots that you want to remove from this repository, and click **Remove selected snapshots**.
4. In the **Remove snapshot** window, confirm the removal of the selected snapshots and click **Remove**.

2.5. UPDATE CUSTOM REPOSITORY AFTER CHANGES

When you make changes to your repository you can trigger a refresh of that repository in the Red Hat Hybrid Cloud Console.

Prerequisites

- You have a RHEL subscription.
- You have administrator access to the Red Hat Hybrid Cloud Console web user interface or **repository administrator** role.
- You updated your custom repository.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Content → Repositories**. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console.
2. Find a repository you want to update and click **Introspect Now** in the **Options** menu.
3. The status of that repository changes to **In progress** that indicates the Hybrid Cloud Console is connecting to the repository and checking for changes.
   The Red Hat Hybrid Cloud Console checks the status of the repositories every 24 hours and again every 8 hours if the status check fails.

2.6. REPOSITORY STATUS IN THE RED HAT HYBRID CLOUD CONSOLE

The repository status shows if the repository is available. The Red Hat Hybrid Cloud Console checks the repository status periodically and can change it. The following table describes the repository status in the Red Hat Hybrid Cloud Console.

Table 2.1. Repository status

Status	Description
Valid	The Red Hat Hybrid Cloud Console has validated the repository and you can use it.
Invalid	The Red Hat Hybrid Cloud Console never validated this repository. You cannot use it.
Unavailable	The repository was valid at least once. The Red Hat Hybrid Console cannot reach this repository at the moment. You cannot use it.
In progress	The repository validation is in progress.

CHAPTER 3. CREATE BLUEPRINTS AND BLUEPRINT IMAGES

An image blueprint is a persistent definition of the required image customizations. You can create multiple builds from it, without having to configure the blueprint for each image build.

You can edit, rebuild, delete, and save a blueprint to keep rebuilding images from it. You can define and manage, by editing or deleting a blueprint, and to keep rebuilding it, removing the need to configure the build each time. When you rebuild a blueprint, all targets specified in the blueprint are rebuilt. The blueprint groups the images that were built from it, so that you can have access to all the related images when dealing with large amounts of images.

The blueprints are persistent and you can manage their customizations. Even if the resulting builds, and images from those builds have different content versions, the customizations stored in that blueprint are always applied across all builds resulting from this blueprint.

When creating an image from the blueprint, unless you define a name to the image, it takes the name of the parent blueprint.

3.1. UNDERSTAND PACKAGE MODE AND IMAGE MODE FOR RED HAT LIGHTSPEED IMAGES

You can build images by using RHEL Lightspeed image builder in traditional package mode or container-native image mode. Understanding the differences between these two modes determines how you manage, update, and deploy your operating system across public clouds and virtualization platforms.

When building a RHEL image, the Image type selection dictates the underlying architecture of how software is delivered and maintained on the lifecycle of the deployed instance.

Package mode (traditional)

Package mode builds a standard RHEL operating system composed of individual RPM packages.

- How it works The image builder resolves dependencies and installs RPMs into a disk image. After you deploy the image, you can manage the system by using package managers such as **dnf**.
- Lifecycle The updates are applied package-by-package. This means two systems that you deployed from the same original image can drift over time as different packages are updated or configuration files are modified independently.

Image mode (container-native)

Image mode treats the entire operating system as a single, immutable container image.

- How it works Instead of managing individual RPMs on the live system, the operating system is packaged as a standard Open Container Initiative (OCI) container image. It uses the **bootc** technology to boot the container image directly on a bare metal environment or a virtual machine.
- Lifecycle The operating system is updated atomically as a single unit by pulling a new version of the container image. If an update fails, the system rolls back to the previous stable version, eliminating configuration drift.

Table 3.1. Comparison between package mode and image mode

Feature	Package Mode	Image mode
Management Unit	Individual RPM packages (**.rpm**)	Entire operating system as a Container Image (OCI format)
Immutability	Mutable: you can add or remove packages, which can be post-deployment	Immutable: read-only root filesystem, changes cleared on reboot
Update Mechanism	Standard package updates by **dnf update**	Atomic updates by pulling and switching to a new container tag
Drift Risk	High because systems can diverge over time	Zero, because all nodes running the same tag are identical

Target environments and deployment use cases

Depending on your use case, you can compile both modes into various target environments. Use the following guidelines to decide where and how to deploy them.

1. Public cloud: AWS, Google Cloud, Microsoft Azure
   You can deliver both modes directly to major cloud providers, but their operational use cases differ:

   - Use package mode if you are deploying traditional multitenant workloads, legacy enterprise software that expects a standard file system, or when you rely on third-party agents that must inject code into standard directories post-boot.

   - Use image mode if you are building cloud-native, auto-scaling applications, where instances are spun up quickly, remain identical, and are replaced rather than patched.

2. Virtualization - guest image (**.qcow2**)
   Use the **.qcow2** format for private clouds and hypervisors such as QEMU or KVM, Red Hat OpenShift Virtualization, or Nutanix.

   - Use package mode if you need a golden primary image for internal corporate VMs that internal IT teams will patch monthly using Red Hat Satellite.

   - Use image mode if you want to use GitOps pipelines to manage your virtualization infrastructure, by using a CI/CD pipeline to rebuild the operating system container image and push it to a private registry for your hypervisors to pull.

3. Container installer (**.iso**)
   Use the **.iso** target format for bare metal deployments, edge computing, or disconnected environments.

   - Use package mode to perform a traditional network or Kickstart installation to a local disk where the machine requires a persistent, uniquely tailored configuration from day one.

   - Use image mode to deploy to edge locations. The container installer **.iso** provisions the hardware with a minimal boot loader that immediately hooks into an immutable operating system container image, ensuring zero-touch maintenance and reliable remote rollbacks.

3.2. CREATE A BLUEPRINT

You can create a blueprint with a name, specify the packages that you want to install and define other customizations. You can build images from this blueprint, and the resulting images have all customizations that you specified in this blueprint.



NOTE

You do not need specific roles to interact with the service by API or Service Account. Standard authentication is enough for general interaction with the image builder API.

If you customize your blueprint or image build process with custom repositories or content sources, you must have the Service Account assigned to a User Access group that has the "Repositories administrator" or the "Repositories viewer" role.

Prerequisites

- You have a RHEL subscription.
- You have Organization Administrator permissions.
- Optional: You have the Activation key for the RHEL system. For details, see Creating an activation key.

Procedure

1. In the Hybrid Cloud Console, navigate to **Inventory → Image Builder**.
   The **Lightspeed Images** dashboard appears.
2. Click **Create image blueprint**.
   The **Create image blueprint** wizard opens.
3. On the **Base settings** page, enter the following information:
   a. In the **Details** section, the console fills a blueprint name in the ***<distribution>-<architecture>-<datetimeString>*** format. You can enter a name for the blueprint, and the system checks for validity and duplicates against already existing blueprint names.
   b. Optional: Add a description for the blueprint.
   c. In the **Image output** section, select the following information:
      i. **Image type**: Choose between the package mode and the image mode.
      ii. **Release**: Select the needed release version.



IMPORTANT

The image builder builds the latest minor version of the selected release unless you use a content template with a specific release stream.

      iii. **Architecture**: Select the architecture that you want to use. The Lightspeed image builder builds the latest minor version of the selected release.

iv. **Target environments**: Select one or more environments that you want to use.

4. In the **Register** section, configure the following details:
   - **Automatically register to Red Hat Hybrid Cloud Console and enable advanced capabilities**
   - **Register later**
   - **Register for a Satellite or Capsule server**
   - **Ansible Automation Platform**
     For more details, see Optional blueprint customizations .
5. In the **Enable repeatable build** section, choose one of the following options:
   - **Disable repeatable build**
   - **Enable repeatable build**
   - **Use a content template**
6. In the **Security** section, select one of the available compliances. For more details, see Optional blueprint customizations. Click **Next**.
7. Optional: On the **Repositories and packages** page, select the required repositories and packages. Click **Next**.
8. Optional: On the **Advanced settings** page, you can further customize your image. For a complete list, see Optional blueprint customizations . .
9. On the **Review** page, verify the details about the image creation, and from the menu:[Create blueprint] drop-down menu, select one of the following options:
   - **Create blueprint**
   - **Create blueprint and build image(s)**
   - **Create the blueprint, save the customizations to it, and build images for the target environment or environments you choose**
     The system verifies the build manifest of the image. After it reaches 100%, the image is displayed in the build queue. The Lightspeed image builder service starts to compose the RHEL image for the selected architecture. After the image build is ready, you can see the images related to the parent blueprint in the **Images** dashboard.

3.3. OPTIONAL BLUEPRINT CUSTOMIZATIONS

To optimize runtime configuration of your image, you can customize and preconfigure your blueprint with optional settings.

For example, you can customize the following settings:

- Automatically register your system.
- Apply your custom compliance policies.
- Configure the file system and partitioning.

- Pre-install additional packages from your custom repositories.
- Add your custom action script for the initial boot of the image.

The additional customizations help you to create your ready to deploy images with all the settings already in place.

Table 3.2. Optional customizations

Customization	Options
Register	**Organization ID** Copy your Organization ID to use it during activation key with command line registration. **Automatically register and enable advanced capabilities** This default option automatically populates the activation key, which you select in the **Activation key** field. While the activation key is loading, the **Next** button gets temporarily disabled. **Register later** Select this option to register your system after you launch the image. **Register to a Satellite or Capsule** To register your system, enter the registration command into the **Satellite registration command** field, and provide the **Certificate authority (CA)**. For more details on how to register a system with Satellite, see Registering a host.
Security	**Enable FIPS mode** Toggle this option to enable Federal Information Processing Standards (FIPS) 140-2 compliant cryptographic algorithms. **Use a custom compliance policy** Select this option to set up your custom policy from the drop-down menu. The images you build with the image builder will automatically appear under the Security Content Automation Protocol (SCAP) policy of the system. You must have an existing set of policies to select from. See Creating new SCAP policies on how to create your policy. If you have not yet created your set of policies, click **Manage Red Hat Lightspeed compliance** to create the policy, and refresh the Console to see the policy that you created under the **Lightspeed compliance** drop-down menu. The system is attached to the customized policy. After you create the image, the system will show up under the compliance system. Note, that this option is available only in the **Preview** mode. **Use a default OpenSCAP profile** Select this option to set up a default OpenSCAP profile from the drop-down menu.

Customization	Options
File system configuration	**Use automatic partitioning** Select this option to enable the automatic partitioning for your image. **Manually configure partitions** Select this option to manually configure the file system for your image. **Advanced disk partitioning** Select this option to define a complete partition table for your image, including an LVM volume group. Note, if you select the **Bare metal** as your target environment, you must configure the file system manually during the installation process.
Repeatable build	**Disable repeatable build** The image uses the latest state of the repository. Choose this option to let the images have the latest content from the repositories. **Enable repeatable build** The image selects a specific date of snapshot creation. Select this option to let the images have the static state of the repository based on a date you specify. **Use a content template** You specify the content template for the image you want to build. Select this option if you plan to centrally manage your systems patching. For details about content templates, see Managing content templates. If you choose to register the system to Lightspeed, it will be automatically associated with the template at the registration time. Note, this option is available only in the **Preview** mode.
Custom repositories	Select the custom repository to add it to your image.
Additional packages	Search and select additional packages to pre-install them to your image.
Groups and users	**Groups** In the **Group name** field add a group. The system automatically assigns a group ID between 1000 and 60000. To add more groups, click **Add group**. **Users** In the **Username** field, add a user, and optionally, assign a password, SSH key, groups, and an **Admin** role to this user. To add more users, click**Add a user** and enter information for those users. Verify that the user group exists when assigning to a user.
Time	**Timezone** From the drop-down menu, select a time zone for your image. **NTP servers** Add the NTP server you need to your image by adding address to the field and click **+**. You can add multiple NTP servers.

Customization	Options
Locale	**Languages** From the drop-down menu, select the language for your image. For example: "*es_NI.UTF-8*". **Keyboard** From the drop-down menu, select the keyboard layout for your image. For example: "*amiga-us*".
Hostname	Select a custom hostname for your image in the **Hostname** field.
Kernel	**Name** From the drop-down menu, select the package for the kernel to use in the image. **Append** You can configure the kernel booting using this option. Enter the kernel arguments into the **Append** field and click **+**.
Firewall	**Ports** Add the port numbers that you want the firewall to manage into the **Ports** field and click **+**. **Enabled service** Add the services that are enabled with the firewall to the **Enabled services** field and click **+**. **Disabled services** Add the services that are disabled with the firewall to the **Disabled services** field and click **+**.
Systemd services	**Enabled services** Add the **systemd** service that you want to enable on your system to the field and click **+**. You can add multiple **systemd** services. **Disabled services** Add the **systemd** service that you want to disable on your system to the field and click **+**. You can add multiple **systemd** services. **Masked services** Add the **systemd** service that you want to disable permanently on your system to the field and click **+**. You can add multiple **systemd** services. For more information about **systemd** services, see Managing systemd.

Customization	Options
Ansible Automation Platform	**Ansible Callback URL** Enter a provisioning callback URL for the job template that you want to run at the launch of your image. To generate the link, you must select the **Provisioning Callbacks** in your Ansible Automation Platform. For more details about job templates, see Creating a job template. **Host Config Key** Enter a host config key for the job template. You can copy it from the **Host config key** on the job template page in your Ansible Automation Platform. **Certificate Authority (CA) for Ansible Controller** Upload a Certificate Authority (CA) for Ansible Controller. For more details about playbooks in Ansible Automation Platform, see Playbooks in Ansible Automation Platform.
First boot configuration	Add your custom script that will run during the first boot of the image. It is useful when you want to enable your custom services or run an Ansible Playbook at the boot time of the image. **Browse** Include a file with an action script. **Start from scratch** Write your action script in the **</> SHELL** field. You must start your script with a shebang, for example the **#!/bin/sh** for Bash shell. The first boot feature supports Python, Bash and YAML syntax.

3.4. COMPLIANCE INTEGRATION IN THE RED HAT ENTERPRISE LINUX LIGHTSPEED IMAGE BUILDER

With the Red Hat Enterprise Linux Lightspeed image builder, you can select base profiles. Each profile includes a predetermined set of rules that are applied to the image. You cannot change the base profile rules.

You can tailor the compliance policies by creating a policy and decide which rules to apply to your image. When you select a compliance policy, Lightspeed image builder adds information about the packages and services that are included and enabled to meet the policy requirements.

When you apply a compliance policy in the Lightspeed image builder, the tool automatically selects and enables the specific packages and system services required to meet that compliance policy.

To apply a compliance policy, your system must already be online and registered with a RHEL subscription with a compliance policy to be able to create compliance policies.

Lightspeed images always builds for the latest minor version of an operating system. But compliance policies are version-specific, leading to issues if a policy does not have a definition for the latest minor version.

3.5. CREATE AND APPLY A COMPLIANCE POLICY TO A CUSTOM IMAGE

Build a custom image that includes a predefined OpenSCAP compliance policy by using Lightspeed image builder.

Prerequisites

- You have a RHEL subscription.
- You must have a system online and registered to be able to create compliance policies.

Procedure

1. Create a compliance policy:
   a. Access the Hybrid Cloud Console, and navigate to **Security → Compliance → SCAP Policies**.
   b. Create the new compliance policy based on the systems you have registered. For more details, see Creating new compliance policies .
   c. Optional: Select or clear rules, based on your requirements.
2. Create a blueprint in the Hybrid Cloud Console by following the Creating a blueprint procedure.
   a. On the **Compliance** step you can choose either from an **Lightspeed Compliance Policy** or **OpenSCAP** profiles. If you have no Lightspeed compliance policies defined, the Hybrid Cloud Console prompts you for the **OpenSCAP** profiles by default.
3. To apply the **Compliance Policy** that you created, navigate to the **Content** section and select **OpenSCAP compliance**.
   a. From the drop-down menu, select the policy you created.
   Lightspeed image builder automatically adds required packages and enables services in the blueprint configuration.
4. Complete the steps for any optional customizations for your image. See Optional blueprint customizations.
5. Review the blueprint details and start to build your image.

Troubleshooting

- Fixing a failing build related to a version mismatch:
  You might see a warning message that the build has failed because a definition for the operating system minor version could not be found in the compliance policy. This happens because the selected compliance policy was created for an older operating system minor version. For example, the policy was created for`RHEL-9.4` and lacks support for the latest minor version, for example, **RHEL-9.6** that Lightspeed image builder is trying to build.

  To work around this issue, follow the steps:

  - Ensure that your registered system is running the exact latest minor version of the target system.

    - Access the Compliance service and edit the policy that you want to use.
    - Update the Compliance policy by attaching your running system.

- Fixing errors related to blueprint:
  If you remove packages marked as required by OpenSCAP, an error pop-up message displays with a **fix errors automatically** button that re-adds those packages.
    - Click the button to update the blueprint.

3.6. EDIT AN EXISTING BLUEPRINT

You can edit a blueprint. For example, to include an extra package. After you finish the blueprint editing, all the images related to the parent blueprint are rebuilt and updated with the new package.

Prerequisites

- You have created a blueprint.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The Lightspeed Images dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Search for the blueprint that you want to edit. You can use the **Search** field to search for the blueprint name.
3. Click the blueprint that you want to edit.
4. Click **Edit blueprint**. The Hybrid Cloud Console redirects you to the **Review** page.
5. From the navigation menu, select the section that you want to edit.
6. After making the changes, click the **Review** page.
7. Click **Save changes to the blueprint**.
   The images related to the parent blueprint that you modified will be rebuilt and updated with the new changes.

3.7. REBUILD AN EXISTING BLUEPRINT

When you edit your blueprint, it creates a new version of itself. That impacts the images that are related to that blueprint, making them out of sync with the blueprint. To ensure you have the latest updates available on your parent blueprint and the related images, edit and rebuild your blueprint.

This action updates all the packages specified in the blueprint and rebuild the related images with the updated packages.

Prerequisites

- You have created a blueprint.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Images**. The Lightspeed Images dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Search for the blueprint that you want to edit. You can use the **Search** field to search for the blueprint name.
3. Click on the blueprint that you want to edit.
4. Click **Edit blueprint**. The Hybrid Cloud Console redirects you to the **Review** page.
5. Make the changes to the blueprint and select one of the options:

   Save

   Save the customizations for your blueprint.

   Save and build image

   Save the customizations for your blueprint and build images from the target environments that you chose.

3.8. IMPORT A BLUEPRINT

You can import a blueprint from an external source to the Hybrid Cloud Console.



IMPORTANT

The **Import** feature is only available in the **Preview** mode.

Prerequisites

- You have created a blueprint in the JSON format.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The Lightspeed Images dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Click **Import**. The **Import pipeline** window opens.
3. Optional: To add custom repositories from the blueprint to your repositories, select the **Import missing custom repositories after file upload** checkbox.



WARNING

If the blueprint contains missing repositories, the images that you create from this blueprint might not work as expected.

4. Upload the blueprint that you want to import either drag and drop or by selecting the file from your local storage.

5. Click **Review and finish**.

3.9. DOWNLOAD A BLUEPRINT

You can export a blueprint that you created in the Hybrid Cloud Console by downloading it in the JSON format.



IMPORTANT

The ability to download a blueprint is only available in the "Preview" mode.

Prerequisites

- You have created a blueprint.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The Lightspeed Images dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Search for the blueprint that you want to download. You can use the Search field to search for the blueprint name.
3. Click the blueprint that you want to download.
4. From the **More options** menu, select the "Download blueprint" option.
   This saves the blueprint as a file in the **.json** format to the local storage that you define in your web browser.

3.10. DELETE A BLUEPRINT

If you no longer need a blueprint, you can delete it. All the images related to this blueprint will also be deleted.

Prerequisites

- You have created a blueprint.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The Lightspeed Images dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Search for the blueprint that you want to delete. You can use the Search field to search for the blueprint name.
3. Click the blueprint that you want to delete.
4. From the **More options** menu, select the "Delete blueprint" option.
5. Confirm the deletion by clicking **Delete**.

CHAPTER 4. BUILD AND MANAGE CUSTOMIZED RHEL IMAGES

You can use a blueprint to build customized RHEL images for a variety of deployment types by using Lightspeed Images. You can build Conventional (RPM-DNF) images.

- You must upload the VMDK customized images to VMWare vSphere, deploy it there, and then you can create a VM.
- For the Guest image (**.qcow2**), and Installer (**.iso**), you can download these images and deploy them directly to virtual machines.
- You can use Lightspeed image builder to build images for the **x86_64** and the **aarch64** architectures.

4.1. ABOUT BUILDING CUSTOMIZED IMAGES

You can build **Conventional (RPM-DNF)** images from a blueprint. You can manage or change the system software by using the DNF package manager and updated RPM packages.

The Red Hat Hybrid Cloud Console saves the image artifacts for 14 days and expires after that. To avoid losing the image, transfer the image to your account before the expiration date. If an image has already expired, you can also re-create the exact image based on an existing blueprint to reuse the previous configuration.

Additional resources

- Learn about image builder API to automate your image building tasks

4.2. BUILD A CUSTOMIZED RHEL SYSTEM IMAGE

Create customized RHEL system images from a blueprint by using Lightspeed Images, and deploy the images on your target environment.

Prerequisites

- You have created a blueprint. See Creating blueprints and blueprint images .

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The list of existing blueprints appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Select the blueprint that you want to build an image from.
   a. Optionally, select the blueprint version from the drop-down menu.
3. Click **Build images**. A pop-up alert informs that the image is being built.
   After the image process status is **Ready**, you can use it in your deployments.

4.3. ACCESS YOUR CUSTOMIZED RHEL IMAGE FOR AWS FROM YOUR ACCOUNT

After you built your image, uploaded it to AWS, and the cloud registration process status is **Ready**, you can access the image that you created and shared with your **AWS EC2** account.

The shared image expires within 14 days. To permanently access your image, copy the image to your own AWS account.

Prerequisites

- You have access to your AWS Management Console.

Procedure

1. Access your AWS account and navigate to **Service → EC2**.
2. In the navigation bar, verify if you are under the correct region: **us-east-1**.
3. Click **Images**, and choose **AMIs**. The dashboard with the **Owned by me** images opens.
4. From the drop-down menu, choose **Private images**.
   You can see the image successfully shared with the **AWS** account you specified.

4.4. LAUNCH YOUR CUSTOMIZED RHEL SYSTEM IMAGE FOR AWS FROM YOUR AWS EC2

Launch the image that you shared with your **AWS** account to the **Amazon Elastic Compute Cloud(Amazon EC2)** compute platform.

Prerequisites

- You have access to your customized image on **AWS**. See Accessing your customized RHEL system image for AWS from your account.

Procedure

1. From the drop-down menu, under **Private images**, locate the image that you shared to the AWS account you specified.
2. Select the image you want to launch.
3. On the top of the panel, click **Launch**. The AWS console redirects you to the **Choose an Instance Type** window.
4. Choose the instance type according to the resources you need to launch your image. **Click Review and Launch**.
5. Review your instance launch details. You can edit each section, such as **Security**, **Storage**, for example, if you need to make any changes. After you finish the review, click **Launch**.
6. To launch the instance, you must select a public key to access it.
   Create a new key pair in EC2 and attach it to the new instance.
   a. From the drop-down menu list, select **Create a new key pair**.
   b. Enter the name to the new key pair. It generates a new key pair.

c. Click **Download Key Pair** to save the new key pair on your local system.

7. Then, you can click **Launch Instance** to launch your instance.
   You can check the status of the instance, it shows as **Initializing**.

8. After the instance status is **running**, the **Connect** button turns available.

9. Click **Connect**. A pop-up window appears with instructions on how to connect by using SSH.

   a. Select the preferred connection method to **A standalone SSH client** and open a terminal.

   b. In the location where you store your private key, make sure that your key is publicly viewable for SSH to work. To do so, run the command:

   ```
   $ chmod 400 <your-instance-name.pem>
   ```

   c. Connect to your instance by using its Public DNS:

   ```
   $ ssh -i "<your-instance-name.pem> ec2-user@<your-instance-IP-address>"
   ```

   d. Type **yes** to confirm that you want to continue connecting.
   As a result, you are connected to your instance over SSH.

Verification

- From a terminal, check if you are able to perform any action while connected to your instance by using SSH.

4.5. COPY YOUR CUSTOMIZED RHEL SYSTEM IMAGE FOR AWS TO A DIFFERENT REGION ON YOUR AWS EC2

You can copy the image you successfully shared with the **Amazon Web Services EC2** to your own account. Doing so, you grant that the image you shared and copied is available until you delete it, instead of expiring after some time. To copy your image to your own account, follow the steps:

Prerequisites

- You have access to your customized image on AWS.

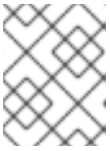
Procedure

1. From the list of **Private images**, select the image you want to copy.

2. On the top of the panel, click **Actions**.

3. From the drop-down menu, choose **Copy AMI**. A window opens.

4. Choose the **Destination region** and click **Copy AMI**.
   After the copying process is complete, you get the new **AMI ID**. You can launch a new instance in the new region.



NOTE

When you copy an image to a different region, it results in a separate and new **AMI** in the destination region, with a unique **AMI ID**.

4.6. AUTHORIZE IMAGE BUILDER TO PUSH IMAGES TO MICROSOFT AZURE CLOUD

You must authorize Lightspeed image builder to push images to the Microsoft Azure cloud. This is a one-time action. The following are high-level steps:

- Configure Lightspeed Images as an authorized application for your **tenant GUID**
- Give the role of **Contributor** to at least one resource group of the authorized application.
  To authorize image builder as an authorized application, follow the steps:

Prerequisites

- You have an existing **Resource Group** in Microsoft Azure portal.
- You have the **User Access Administrator** role rights.
- Your Microsoft Azure subscription has **Microsoft.Storage** and **Microsoft.Compute** as a resource provider.

Procedure

1. Access the Hybrid Cloud Console, and navigate to **Inventory → Image Builder**. The Lightspeed image builder dashboard appears. For details on how to access the Red Hat Hybrid Cloud Console, see Getting started with the Red Hat Hybrid Cloud Console .
2. Click **Create image blueprint**. The **Image output** wizard opens.
3. On the **Image output** page, complete the following steps:
   a. From the **Release** list, select the release that you want to use.
   b. From the **Select target environments** option, select **Microsoft Azure**.
   Click **Next**.
4. On the **Target Environment - Microsoft Azure**window, to add image builder as an authorized application, select one of the following **share method** options:
   a. **Use an account configured from Sources**
      i. From the **Source name** drop-down menu, select the source that you previously configured. See Adding a Microsoft Azure account as a cloud integration.
5. The **Azure tenant GUID**, the **Subscription ID**, and the **Resource group** are automatically completed, and the **Authorize image builder** button becomes available.
   Image builder checks if your **Tenant GUID** is correctly formatted and the **Authorize image builder** button becomes available.
   a. **Manually enter the account information**
      i. Enter your **Azure Tenant GUID**.

Image builder checks if your **Tenant GUID** is correctly formatted and the **Authorize image builder** button becomes available.

b. One time action: Click **Authorize image builder** to authorize image builder to push images to the Microsoft Azure cloud.
This redirects you to the Microsoft Azure portal.

   i. Login with your credentials.

   ii. Click **Accept** the **Permission requested**. Note that, if you already went through the authentication process before, you will not see the **Permission requested**. It is already granted.

c. Confirm that image builder is authorized for your tenant.

   i. In the **search** bar, search for **Azure Active Directory**.

   ii. From the **Services** menu, click **Microsoft Entra ID**, from the left menu. The **Azure Active Directory** page opens.

   iii. Search for Lightspeed image builder and confirm it is authorized.

   iv. In the **Azure Active Directory**, from the **Services** list, select **Enterprise applications**.

   v. In the **Enterprise applications** page, from the **Manage list** menu, click **All applications**. You can see **Red Hat Image Builder** is authorized in the Microsoft Azure cloud.

d. Add the **Red Hat Image Builder** as a contributor to your **Resource Group**.

   i. In the search bar, type **Resource Groups** and select the first entry under **Services**. This redirects you to the **Resource Groups** dashboard.

   ii. Search and select your **Resource Group** by name.

   iii. On the lateral menu, click **Access control (IAM)** to add a permission to the **Red Hat Image Builder** application to access your resource group.

   iv. From the menu, click the tab **Role assignments**.

   v. Click **+Add**.

   vi. From the dropdown menu, choose **Add role assignment**. A menu opens on the left side.

   Select role

   Assign the **Contributor** role.

   Assign access to

   Select the option **Assign access to user, group, and service principal**

   Members

   Click **+Select members** and type **Red Hat** in the search bar. Click **enter**.

   Select

   Red Hat Image Builder application.
   The Red Hat image builder application is now authorized to push images to the Microsoft Azure cloud.



NOTE

The Red Hat image builder application can locate resources only when the account administrator adds the shared application as a contributor under the **IAM** section of the resource group.

Verification

- From the menu, click the tab **Role assignments**.
  You can see Red Hat image builder set as a **Contributor** of the **Resource Group** you selected.

Additional resources

- Manage Microsoft Azure Resource Manager resources group by using the Microsoft Azure portal

4.7. ACCESS YOUR CUSTOMIZED RHEL SYSTEM IMAGE FROM YOUR MICROSOFT AZURE ACCOUNT

After finishing to build and upload the image, and the cloud registration process status is **Ready**, you can access the **Azure Disk Image** from your **Microsoft Azure** account.

Prerequisites

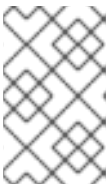
- You have access to your Microsoft Azure dashboard.

Procedure

- Access your **Microsoft Azure** dashboard and navigate to the **Resource group** page.

Verification

1. After you access your Microsoft Azure Account, you can see that the image successfully shared with the resource group account you specified.



NOTE

If the image is not visible there, you might have issues with the upload process. Return to the Lightspeed image builder dashboard and check if the image is **Ready**.

4.8. CREATE A VM INSTANCE BY USING YOUR GOOGLE CLOUD IMAGE

After you build and upload the image, and the cloud registration process status is **Ready**, you can create a Virtual Machine (VM) instance by using the Google Cloud image.

Prerequisites

- You have the universally unique identifier (UUID) of the image you created.

- You have access to the image-builder service API endpoint.
- You have access to your project details at Google Cloud.
- You can access Google Cloud Shell from your browser.

Procedure

1. From the Lightspeed image builder dashboard, copy the image **UUID** of the image that you created.
2. Access /composes/{composeId} API endpoint.
3. Click **Try it Out** to activate the **composeId** string path.
4. Enter the **UUID** into the **composes/{composeId}** field in the API endpoint.
5. Click **Execute**. The API endpoint generates a response in the **Response body**, for example:

```
{
  "image_status": {
    "status": "success",
    "upload_status": {
      "options": {
        "image_name": "composer-api-03f0e19c-0050-4c8a-a69e-88790219b086",
        "project_id": "red-hat-image-builder"
      },
      "status": "success",
      "type": "gcp"
    }
  }
}
```

6. From the **Response body** field, copy the ***image_name*** and ***project_id*** to access the image from the Google Cloud environment.
7. From your browser, access Google Cloud Shell and set your Google Cloud **Project ID** as the default project. You can find the Product ID of your project by accessing the Google Cloud dashboard.

```
$ gcloud config set project PROJECT_ID
```

8. In the **Authorize Cloud Shell** window prompt, click **Authorize** to allow this and future calls that require your credentials.
9. Create a VM instance with the image by using the **gcloud** command in the Google Cloud Shell:

```
$ gcloud compute instances create INSTANCE_NAME \
  --image-project PROJECT_ID_FROM_RESPONSE \
  --image IMAGE_NAME \
  --zone GCP_ZONE
```

Where:

- *INSTANCE_NAME* is the name for your instance;

- *PROJECT_ID_FROM_RESPONSE* is the *project_id* generated by **Response body**;
- *IMAGE_NAME* is the *image_name* generated by **Response body**;
- *GCP_ZONE* is the Google Cloud zone in which you will create the instance.

Verification

1. Verify that Compute Engine created the VM:

```
$ gcloud compute instances describe INSTANCE_NAME
```

2. Connect to the VM instance using SSH:

```
$ gcloud compute ssh --project=PROJECT_ID --zone=ZONE INSTANCE_NAME
```

Additional resources

- Google Cloud Shell documentation
- Default region and zone

4.9. COPY THE GOOGLE CLOUD IMAGE TO YOUR PROJECT GROUP

You can create a Virtual Machine (VM) instance by using the Google Cloud image.

Prerequisites

- The universally unique identifier (UUID) of the image you created.
- Access to the Image-builder service API endpoint.
- Access to the Google Cloud Shell from your browser.

Procedure

1. From the Images dashboard, copy the **UUID** image of the image you created.
2. Access /composes/{composeId} API endpoint.
3. Click **Try it Out** to activate the **composeId** string path.
4. Enter the **UUID** into the **composes/{composeId}** field in the API endpoint.
5. Click **Execute**. The API endpoint generates a response in the **Response body**, for example:

```
{
  "image_status": {
    "status": "success",
    "upload_status": {
      "options": {
        "image_name": "composer-api-03f0e19c-0050-4c8a-a69e-88790219b086",
        "project_id": "red-hat-image-builder"
      },
```

```
      "status": "success",
      "type": "gcp"
    }
  }
}
```

6. From the **Response body** field, copy the ***image_name*** and ***project_id*** to access the image from the Google Cloud environment. From the **Response body**:

```
"image_name": "composer-api-03f0e19c-0050-4c8a-a69e-88790219b086",
"project_id": "red-hat-image-builder"
```

7. From your browser, access Google Cloud Shell.

8. Set your Google Cloud **Project ID** as the default Google Cloud Project (GCP). You can find the Product ID of your project by accessing the Google Cloud dashboard.

```
$ gcloud config set project PROJECT_ID
```

9. In the **Authorize Cloud Shell** window prompt, click **Authorize** to allow this and future calls that require your credentials.

10. Copy the image to your project by using the gcloud command:

```
$ gcloud compute images create MY_IMAGE_NAME \
  --source-image-project red-hat-image-builder \
  --source-image IMAGE_NAME
```

Where:

- *MY_IMAGE_NAME* is the name you give to your instance;
- red-hat-image-builder is the *project_id* generated by **Response body**;
- *IMAGE_NAME* is the *image_name* generated by **Response body**;

Verification

Confirm that the image has been successfully copied to your project:

- Using the Google Cloud UI, by accessing the Compute Engine / Images section.
- Using the **gcloud** tool, by running the command in Google Cloud Shell:

```
$ gcloud compute images list --no-standard-images
```

Additional resources

- Google Cloud Shell documentation

4.10. CREATE A DEPLOYMENT-READY IMAGE BY USING IMAGE MODE

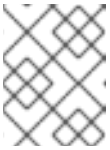
In the RHEL image builder wizard, you can use the **Base settings** step to define the core identity, architecture, and deployment targets for your image. By creating an image mode output, you can manage the operating system as an immutable, containerized artifact.

Prerequisites

- You have a RHEL subscription.
- You have Organization Administrator permissions.
- Optional: You have the activation key for the RHEL system. For more information, see Creating an activation key.
- You have determined your target deployment environment, such as AWS, Microsoft Azure, Google Cloud, or virtualization.

Procedure

1. In the Red Hat Hybrid Cloud Console, navigate to **Inventory → Image Builder**. The Lightspeed Images dashboard appears.
2. Click **Create image blueprint**.
   The Create image blueprint wizard opens.
3. On the **Base settings** page, define the image details by entering a unique name to identify your deployment-ready image, and optionally, a brief description explaining the purpose or contents of the image.
4. Select the **Image output** options and click **Next**:
   - **Image type**: Select **Image mode**.

     NOTE

     RHEL image mode treats the entire operating system as a single, immutable container image with the atomic updates.

   - **Image source**: Select your preferred RHEL version (for example, RHEL 10).
   - **Architecture**: Select the system architecture from the dropdown, for example, **x86_64**.
   - **Target environments**: Select the environments where you want to deploy the image.
5. Depending on the **Target environments** that you choose, you might need to configure additional settings and dependencies in later wizard steps.

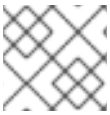
Next steps

- Configure registration and subscription keys.
- Define custom repositories or additional packages if required by your target environment.

4.11. CREATE A NEW IMAGE FROM AN EXISTING BUILD

You can create a new image from an existing customized RHEL image by using Lightspeed Images. The Lightspeed Images re-creates the exact image, with a different UUID, which you can use to identify the image in the Hybrid Cloud Console.

The new image also fetches package updates and refreshes the content with those updates. You can customize this new image to fit your requirements.



NOTE

You can re-create images from failed builds.

Prerequisites

- You created an AWS image with Lightspeed Images.

Procedure

1. From the **Images** dashboard, select the image from which you want to create your customized image.
2. Click the **Node options** menu (▮) and select **Re-create image**. The **Create image** wizard opens.



NOTE

If the image status is **Expired**, click the **Re-create image** button.

   a. Optional: You can customize the new image by using the Navigation panel to open a step and making changes. Click **Next**.

   b. On the **Review** page, click **Create image**.
   The Lightspeed Images dashboard opens. The image build starts to re-create the image and lists the following information:

      - **Image name**
      - **UUID**
      - Cloud target environment
      - Image operating system release
      - Status of the image creation

Verification

- From the **Status** column, check if the image is **Ready**.
- Optional: Click **Image details** to display additional information about the re-created image.

4.12. DOWNLOAD THE JSON COMPOSE REQUEST

If you download the **.json** compose request of your image, you can use the image builder **API** to automate your image building tasks. For example, you can automate customizing the image with extra packages, customizing the partition layout, or embedding an activation key.

Prerequisites

- You created an image with Lightspeed Images.

Procedure

1. From the **Images** table, select the image that you want to download as a **.json** compose request.
2. Click the **Node options** ( ) menu and select **Download compose request (.json)**.
   The **.json** compose request is now saved to your host server. To use the image builder API, see Using hosted image builder via its API .

CHAPTER 5. DEPLOY YOUR CUSTOMIZED IMAGES

After creating customized images for the VMWare vSphere private cloud, you can deploy the images to VMware vSphere. For the Guest image (**.qcow2**), and Installer (**.iso**), you can then download these images and deploy them to virtual machines.



NOTE

To create a VM, you must upload the VMDK customized images to VMWare vSphere and deploy it there. For the Guest image (**.qcow2**), and Installer (**.iso**), you can then download these images and deploy them directly to virtual machines.

5.1. CONFIGURE CREDENTIALS TO ACCESS YOUR DEPLOYED SYSTEMS WITH CLOUD-INIT

You can add a username and password to your image by using the **cloud-init** tool.

The following example shows how to add credentials to a VMware Open Virtual Appliance (OVA) image type created with Lightspeed image builder. You can add credentials to other image types by using this method.

The Open virtualization format (**.ova**) is a **.vmdk** image with additional metadata about the virtual hardware. The **.ova** image contains the **cloud-init** package installed that you can use to provision users by using a **user-data** file, for example.

Instead of sharing your login credentials to a hosted service, use **cloud-init** and **open-vm-tools**, which are installed within the image and enabled by default. For example, you can use it to pass the credentials to the VMware vSphere Cloud Director by using **cloud-init**.

Prerequisites

- You created an image in the **.ova** format by using the Hybrid Cloud Console.

Procedure

1. Access the directory where you downloaded your **.ova** image.
2. Create a file named **metadata.yaml** and add the following information to this file:

```
instance-id: cloud-vm
local-hostname: vmname
```

3. Create a file **userdata.yaml**. Add the following information to the file:

```
#cloud-config
users:
- name: admin
  sudo: "ALL=(ALL) NOPASSWD:ALL"
  ssh_authorized_keys:
  - ssh-rsa AAA...fhHQ== your.email@example.com
```

This file contains the administrator or root user credentials with no password that you can use to access your system after the first boot and create additional users later. The

ssh_authorized_keys field is your SSH public key. You can find your SSH public key in **~/.ssh/id_rsa.pub**.

Next steps

- Deploy your image to vSphere by using the CLI or to the vSphere GUI.

5.2. CONFIGURE CREDENTIALS TO ACCESS YOUR DEPLOYED SYSTEMS WITH API

You can add a username and password to a blueprint by using the Image-builder service API.

The following example shows how to add credentials to a VMware OVA image type created with Lightspeed image builder. You can also use this method to add credentials to other image types.

Prerequisites

- Access to Hybrid Cloud Console.
- OAuth 2.0 authorization.
- You created an access token by using the image builder API. See Generating an access token for the Image Builder API queries.
- You installed the **jq** tool.

Procedure

1. Generate an access token for API queries by following the step in Generating an access token for the Image Builder API queries.



IMPORTANT

Access tokens expire after fifteen (15) minutes. The expiration is included within the JSON response. You must repeat the process of obtaining an access token about every ten (10) minutes to rotate the token before it expires.

2. In your system, create a file in the **.json** format to create an image. The following example shows how to create an up-to-date RHEL 9.4 **ova** image for x86_64 CPU architecture.

```
$ request.json
{
    "image_name": "ova_image_name",
    "distribution": "rhel-94",
    "image_requests": [
        {
            "architecture": "x86_64",
            "image_type": "vsphere-ova",
            "upload_request": {
                "type": "vmdk",
                "options": {}
            }
        }
    ],
```

```
    "customizations": {
        "users": [
            {
                "name": "user-name",
                "ssh_key": "ssh-rsa AAAAB...qfGI+vk",
                "password": "password"
            }
        ]
    }
}
```

3. Send the compose request to the image builder API:

```
$ curl --silent \
    --request POST \
    --header "Authorization: Bearer $access_token" \
    --header "Content-Type: application/json" \
    --data @request.json \
    https://console.redhat.com/api/image-builder/v1/compose
```

If the request is successful, you can see an output similar to the following, which is the image ID:

```
{"id":"fd4ecf3c-f0ce-43dd-9fcc-6ad11208b939"}
```

4. Check the status of the image building:

```
$ curl \
    --silent \
    --header "Authorization: Bearer $access_token" \
    "https://console.redhat.com/api/image-builder/v1/composes/$compose_id" \
  | image_ID.
```

If the request is successful, you can see an output similar to the following, which is the image ID:

```
{"id":"fd4ecf3c-f0ce-43dd-9fcc-6ad11208b939"}
```

You can also check the image building progress by accessing Red Hat Hybrid Cloud Console .

After the image builds, you can see the following output:

If the request is successful, you can see an output similar to the following, which is the image ID:

```
{
    "image_status": {
        "status": "success",
        "upload_status": {
            "options": {
                "url": "https://image-builder-service-production.s3.amazonaws.com/composer-api-
76...-disk.ova?e42..."
            },
            "status": "success",
            "type": "aws.s3"
```

```
        }
    }
}
```

5. After finishing the image creation, download the image.

```
$ curl --location --output vsphere-ova.vmdk  \
    “https://image-builder-service-production.s3.amazonaws.com/composer-api-76...-disk.ova?
e42...”
```

As a result, you have the image saved to your system and can use it.

Next steps

- Deploy your image to vSphere by using the CLI or to the vSphere GUI.

Additional resources

- Getting started with Red Hat APIs
- Image-builder service API
- Transition of Red Hat Hybrid Cloud Console APIs from basic authentication to token-based authentication via service accounts
- Red Hat Lightspeed Information and Resources

5.3. UPLOAD VMDK IMAGES AND CREATE A RHEL VIRTUAL MACHINE IN VSPHERE

After creating and configuring your image, you can deploy it to **VMware vSphere** by using the CLI, and you can create a VM and log in to it.

Prerequisites

- You configured credentials to access your deployed systems by using the **cloud-init** tool. See Configuring credentials to access your deployed systems .
- You configured the **govc** VMware CLI tool client with the following values in the environment by setting the following values in the environment:

```
GOVC_URL
GOVC_DATACENTER
GOVC_FOLDER
GOVC_DATASTORE
GOVC_RESOURCE_POOL
GOVC_NETWORK
```

Procedure

1. Export the **metadata.yaml** and **userdata.yaml** files to the environment, compressed with **gzip**, encoded in **base64** as follows. You will use them in further steps.

```
export METADATA=$(gzip -c9 <metadata.yaml | { base64 -w0 2>/dev/null || base64; }) \
USERDATA=$(gzip -c9 <userdata.yaml | { base64 -w0 2>/dev/null || base64; })
```

2. Launch the image on vSphere with the **metadata.yaml** and **userdata.yaml** files:

   a. Import the **.vmdk** image in to vSphere:

   ```
   $ govc import.vmdk ./composer-api.vmdk foldername
   ```

   b. Create the VM in vSphere without powering it on:

   ```
   govc vm.create \
   -net.adapter=vmxnet3 \
   -m=4096 -c=2 -g=rhel8_64Guest \
   -firmware=bios -disk="foldername/composer-api.vmdk" \
   -disk.controller=ide -on=false \
    vmname
   ```

   c. Change the VM to add **ExtraConfig** variables, the **cloud-init** config:

   ```
   govc vm.change -vm vmname \
   -e guestinfo.metadata="${METADATA}" \
   -e guestinfo.metadata.encoding="gzip+base64" \
   -e guestinfo.userdata="${USERDATA}" \
   -e guestinfo.userdata.encoding="gzip+base64"
   ```

   d. Power-on the VM:

   ```
   govc vm.power -on vmname
   ```

   e. Retrieve the VM IP address:

   ```
   HOST=$(govc vm.ip vmname)
   ```

   f. Use SSH to log in to the VM, using the user-data specified in **cloud-init** file configuration:

   ```
   $ ssh admin@HOST
   ```

Additional resources

- The govc documentation
- VMware - cloud init 22.2 documentation

5.4. DEPLOY OVA VMDK IMAGES TO THE VSPHERE GUI

After creating your **.vmdk** image in the **open virtualization** format (**.ova**), you can deploy it to **VMware vSphere** by using the vSphere GUI client. It will create a virtual machine (VM), which you can customize further before booting.

Prerequisite

- You logged in to the vSphere UI in a browser.
- You downloaded your (**.ova**) image.

Procedure

1. In the vSphere Client, from the **Actions** menu, select **Deploy OVF Template**.
2. On the **Deploy OVF Template** page, complete the settings for each configuration option and click **Next**.
3. To deploy the **.ova** image, click **Finish**.
   After the image deployment is complete, you have a new VM from the **.ova** image.
4. In the deployed image page, perform the following steps:
   a. From the **Actions** menu, select **Edit Setting**.
   b. On the **Virtual Hardware** tab, configure resources such as CPU, memory, add a new network adapter, among others of your choice.
      i. On the **CD/DVD drive 1** option, attach a CD or DVD Drive that contains a **cloud-init.iso**, to provision a user on startup.
      The VM is now ready to boot with the username and password from the **cloud-init.iso** file.

Additional resources

- Deploy an OVF or OVA Template
- The govc documentation
- VMware - cloud init 22.2 documentation

5.5. CREATE A VIRTUAL MACHINE FROM THE CUSTOMIZED RHEL GUEST SYSTEM IMAGE

You can create a virtual machine (VM) from the **QCOW2** image that you created by using Lightspeed Images.

Prerequisites

- You created and downloaded a **QCOW2** image by using Lightspeed Images.

Procedure

1. Access the directory where you downloaded your **QCOW2** image.
2. Create a file named **meta-data**. Add the following information to this file:

```
instance-id: nocloud
local-hostname: vmname
```

3. Create a file named **user-data**. Add the following information to the file:

```
#cloud-config
user: admin
password: password
chpasswd: {expire: False}
ssh_pwauth: True
ssh_authorized_keys:
  - ssh-rsa AAA...fhHQ== your.email@example.com
```

- **ssh_authorized_keys** is your SSH public key. You can find your SSH public key in ~/**.ssh/id_rsa.pub**.

4. Use the **genisoimage** command to create an ISO image that includes the **user-data** and **meta-data** files.

```
# genisoimage -output cloud-init.iso -volid cidata -joliet -rock user-data meta-data

I: -input-charset not specified, using utf-8 (detected in locale settings)
Total translation table size: 0
Total rockridge attributes bytes: 331
Total directory bytes: 0
Path table size(bytes): 10
Max brk space used 0
183 extents written (0 MB)
```

5. Create a new VM from the KVM Guest Image using the **virt-install** command. Include the ISO image you created on step 4 as an attachment to the VM image.

```
# virt-install \
   --memory 4096 \
   --vcpus 4 \
   --name myvm \
   --disk composer-api.qcow2,device=disk,bus=virtio,format=qcow2 \
   --disk cloud-init.iso,device=cdrom \
   --os-variant rhel1-latest \
   --virt-type kvm \
   --graphics none \
   --import
```

Where,

- --graphics none - indicates that it is a headless RHEL Virtual Machine.
- --vcpus 4 - indicates that it uses 4 virtual CPUs.
- --memory 4096 - indicates that it uses 4096 MB RAM.

6. The VM installation starts:

```
Starting install...
Connected to domain myvm
...
[  OK  ] Started Execute cloud user/final scripts.
[  OK  ] Reached target Cloud-init target.
```

```
Red Hat Enterprise Linux 1-latest (Ootpa)
Kernel 4.18.0-221.el8.x86_64 on an x86_64
```

Additional resources

- Creating virtual machines using the command-line interface

5.6. INSTALL A CUSTOMIZED RHEL ISO SYSTEM IMAGE TO A BARE-METAL SYSTEM

You can create a virtual machine (VM) from the ISO image that you created using the Lightspeed image builder.

Prerequisites

- You created and downloaded an ISO image by using Lightspeed image builder.
- A 8 GB USB flash drive.

Procedure

1. Access the directory where you downloaded your ISO image.
2. Place the bootable ISO image file on a USB flash drive.
3. Connect the USB flash drive to the port of the computer you want to boot.
4. Boot the ISO image from the USB flash drive.
5. Perform the steps to install the customized bootable ISO image.
   The boot screen shows you the following options:
   - **Install Red Hat Enterprise Linux 1-latest**
   - **Test this media & install Red Hat Enterprise Linux 1-latest**

Additional resources

- Booting the installation

5.7. IMPORT AND RUN QCOW2 IMAGES ON OCI

You can import your customized **.qcow2** image to the Oracle Cloud Infrastructure (OCI). Then, you can launch the customized **.qcow2** image on a virtual machine.

Prerequisites

- You logged in to the Oracle Cloud UI in a browser.
- You downloaded your **.qcow2** image.
- You have copied the **.qcow2** image **Image Link** URL from Lightspeed **Instance** column, in the Images dashboard.

Procedure

1. In the Oracle Cloud UI dashboard, click **Compute > Custom Images**
2. On the **Custom Images** dashboard, click **Import image**.
3. On the **Import image** window, set the following configuration:
    a. Select the **Import from an object storage URL** option.
    b. In the **Object Storage URL** field, paste the URL given by Lightspeed Images into it.
    c. Choose the **QCOW2** image type.
    d. Under **Launch mode**, select the **Paravirtualized mode** option.
4. Click **Import Image**.

Next steps

- After the system finishes importing the image, you can run the customized image in the OCI environment.

CHAPTER 6. LAUNCH CUSTOMIZED RHEL IMAGES TO THE PUBLIC CLOUD PLATFORMS WITH THE CLOUD PROVIDER TOOLS

You can launch your customized RHEL image to the public cloud platforms by using the tools that are available from the cloud providers.

6.1. LAUNCH CUSTOMIZED IMAGES ON AWS WEB CONSOLE

You can launch your customized image on the Amazon Web Service (AWS) by using AWS Management Console.

Prerequisites

- You have already built a custom image for your target environment on AWS.
- You have access to your custom image on AWS.

Procedure

1. In the AWS Management Console, navigate to **Compute → EC2 → Images**.
2. Copy your custom image to your AWS account to create a permanent copy. For details, follow the Copying your customized RHEL system image for AWS to a different region on your AWS EC2 procedure.
3. From the AWS Management Console, launch your image as an instance.

Verification

- From your terminal, connect to your instance with the **ssh** command, and using the **ec2-user** username.

6.2. LAUNCH CUSTOMIZED IMAGES ON MICROSOFT AZURE

You can launch your custom image on the Microsoft Azure by using Microsoft Azure portal.

Prerequisites

- You have already built a custom image for your target environment on Microsoft Azure.
- You have access to your custom image on Microsoft Azure.
- You granted permissions for image builder to push images to Microsoft Azure. For details, see Authorizing image builder to push images to Microsoft Azure Cloud .

Procedure

1. Access the **Images** on the Microsoft Azure portal, and click on your custom image.
2. On the **Overview** dashboard, click **Create VM**.

3. In the Microsoft Azure wizard, configure the settings for the new virtual machine that you want to launch.

4. On the **Review + create** tab, review the information about your image and click **Create**.

Verification

- After the deployment is complete, access **Infrastructure → Virtual machines**, and locate your new VM.

6.3. LAUNCH CUSTOMIZED IMAGES ON GOOGLE CLOUD

You can launch your custom image on the Google Cloud by using the Google Cloud command line interface.

Prerequisites

- You have the **gcloud** cli installed.
- You have already built a custom image for your target environment on Google Cloud.

Procedure

1. Log in to your Google Cloud account from your terminal:

```
$ gcloud auth login
```

2. Set the Google project where you want to create your image:

```
$ gcloud config set project your-project-name
```

3. Create the custom image:

```
$ gcloud compute images create image-name --source-image composer-api-image-builder-uuid --source-image-project red-hat-image-builder
```

Copy the *image-builder-uuid* from the ComposeID API endpoint.

4. Access the Google Cloud Console from your browser, and locate your new image in **Compute Engine → Images**.

5. To launch a new virtual machine from your image, click **Actions → Create Instance**.

Verification

- In the Google Cloud console, navigate to the **VM instances** page. The **Status** of your new instance is running with a green checkmark icon.

Next steps

- Access your instance via **ssh** command from your terminal. You can find the command on the **Details** page of the instance.

CHAPTER 7. IMAGE BUILDER API

The image builder API contains examples for using various types of request. Administrators and developers can use the image builder API to write custom scripts to integrate image builder services with third-party applications.

7.1. GENERATE AN ACCESS TOKEN FOR THE IMAGE BUILDER API QUERIES

The image builder API queries require an active access token. The access token expires after 900 seconds after its creation. To generate the access token, complete these steps.

Prerequisites

- You have a RHEL subscription.
- You have a **service account** in the Red Hat Hybrid Cloud Console. For details, see Creating a service account.

Procedure

- To generate the access token, include the following code in your script in the **Authorization** header with the value **Bearer $ACCESS_TOKEN**:

```
CLIENT_ID="YOUR_CLIENT_ID"
CLIENT_SECRET="YOUR_CLIENT_SECRET"
ACCESS_TOKEN=$( \
    curl -d "client_id=$CLIENT_ID" \
        -d "client_secret=$CLIENT_SECRET" \
        -d "grant_type=client_credentials" \
        "https://sso.redhat.com/auth/realms/redhat-external/protocol/openid-connect/token" \
        -d "scope=api.console" \
    | jq -r .access_token \
)
```

  Replace *YOUR_CLIENT_ID* with the client ID, and *YOUR_CLIENT_SECRET* with the client secret from your **service account**.

Additional resources

- Image Builder API catalogue

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate and prioritize your feedback regarding our documentation. Provide as much detail as possible, so that your request can be quickly addressed.

Prerequisites

- You are logged in to the Red Hat Customer Portal.

Procedure

1. Click the following link: Create Issue.
2. Describe the issue or enhancement in the **Summary** text box.
3. Provide details about the issue or requested enhancement in the **Description** text box.
4. Type your name in the **Reporter** text box.
5. Click the **Create** button.
   This action creates a documentation ticket and routes it to the appropriate documentation team. Thank you for taking the time to provide feedback.

OPENING A SUPPORT CASE AT RED HAT SUPPORT

Create a support case from Red Hat Lightspeed at Red Hat Support by performing the following steps:

Prerequisites

- You are logged in to the Red Hat Customer Portal.

Procedure

1. Access the Red Hat Hybrid Cloud Console :
2. Click **Help ?** and select **My support cases**.
3. In **My support cases** tab, click **Open a support case**.
   You are redirected to the **Customer support** page.
4. From the **Get Support** page, select the type of issue that you want to report and click **Continue**.
5. From the **Summarize** page, perform the following steps:
   a. On the **Summary** field, describe the issue.



NOTE

If **Red Hat Lightspeed** is not auto-selected, you must manually select the product.

   b. From the **Product** drop-down menu, select **Red Hat Lightspeed**.
   c. From the **Version** drop-down menu, select the component you have issues with.
   d. From the **Review** page, click **Submit**.
   A support case is created.